

Compact space.

1. a). Let $\mathcal{C}, \mathcal{C}'$ be topologies on X , with $\mathcal{C} \subseteq \mathcal{C}'$.

If $(X, \mathcal{C}')$ is compact, then $(X, \mathcal{C})$ is compact.

Proof. Let $\mathcal{O} \in \mathcal{C}$ be an open cover of X . Then, since $\mathcal{O} \subseteq \mathcal{C}'$ and $(X, \mathcal{C}')$ compact, $\mathcal{O}$ has finite subcover. $\square$

1. b). If X is Compact T_2 under both $\mathcal{C}$ and $\mathcal{C}'$,

then either $\mathcal{C}$ and $\mathcal{C}'$ are equal or they are Not comparable.

Proof. If $\mathcal{C} = \mathcal{C}'$, then there is nothing to prove. Suppose $\mathcal{C} \neq \mathcal{C}'$.

Enough to show that neither $\mathcal{C} \subseteq \mathcal{C}'$ nor $\mathcal{C}' \subseteq \mathcal{C}$.

Suppose that $\mathcal{C} \subseteq \mathcal{C}'$. Then, we can choose an open $U \in \mathcal{C}' \setminus \mathcal{C}$.

Now, $X \setminus U$ is closed in $\mathcal{C}'$ but not $\mathcal{C}$. Since $\mathcal{C}'$ is Compact T_2 ,

$X \setminus U$ is compact set of $\mathcal{C}'$. Since $\mathcal{C} \subseteq \mathcal{C}'$, for any open cover $\mathcal{O} \subseteq \mathcal{C}$ for $X \setminus U$,

it has finite subcover. Therefore, $X \setminus U$ is compact in $(X, \mathcal{C})$, thus closed.

But, it is contradiction. Similarly, $\mathcal{C}' \not\subseteq \mathcal{C}$.

2. a). Every subspace in Co-finite Topology $(\mathbb{R}, \mathcal{C}_f)$ is compact.

b). In Co-Countable $(\mathbb{R}, \mathcal{C}_c)$, is $[0, 1]$ compact? an

Proof. Let $A \subseteq \mathbb{R}$ be a subset. Let $\mathcal{O} = \{\mathbb{R} \setminus F \mid F \text{ is finite}\}$ be open cover for A . There is,

$$A \subseteq \bigcup_{U \in \mathcal{O}} U \Leftrightarrow \bigcap_{U \in \mathcal{O}} \mathbb{R} \setminus U \subseteq A^c$$

But since $\mathbb{R} \setminus U$ is finite for each $U \in \mathcal{O}$, there are finite number of $U_i, \dots, U_n$ s.t.

$$\bigcap_{i=1}^n \mathbb{R} \setminus U_i \subseteq A^c \Leftrightarrow A \subseteq \bigcup_{i=1}^n U_i$$

Therefore A is Compact. (More specifically, for any distinct finite set F_1 and F_2 , $F_1 \cap F_2$ has number of less than $\min\{\#(F_1), \#(F_2)\}$.)

But, in the Co-Countable, $[0, 1]$ is Not compact, because

Construct sequences; denote $\mathcal{O} \cap [0, 1] = \{U_n \mid n \in \mathbb{N}\}$, and let $\mathcal{U}_n = (\mathbb{R} \setminus \mathbb{Q}) \cup \{x_n\}$

Then, $\bigcup_{n=1}^{\infty} U_n \supseteq [0, 1]$ but it has no finite subcover.

3. Finite Union of Compact Subspace is Compact.

Proof. Let $K_1, \dots, K_n \subseteq X$ be Compact subsets. Let $\mathcal{O} \subseteq \mathcal{P}(X)$ be an open cover of $\bigcup_{i=1}^n K_i$.

Since $\bigcup_{U \in \mathcal{O}} U \supseteq \bigcup_{i=1}^n K_i \supseteq K_j$ ($1 \leq j \leq n$), for each j , $\mathcal{O}$ has a finite subcover for K_j ,

denote $\mathcal{U}_j^1, \dots, \mathcal{U}_j^{r_j} \in \mathcal{O}$. Now, clearly

$$\bigcup_{i=1}^n K_i \subseteq \bigcup_{j=1}^n \bigcup_{m=1}^{r_j} \mathcal{U}_j^m$$

4. Every Compact Subspace in a Metric space is closed bounded.

Give the example that the Converse fails.

Proof. Closed is given by: Compact in T_2 is closed, because

Let X be a Hausdorff space, and $K \subseteq X$ be a Compact subset.

Let $x \in X \setminus K$ be fixed. Now, for each $y \in K$, there exists disjoint open $\mathcal{U}_y^x, \mathcal{U}_y$ s.t.

$$x \in \mathcal{U}_y^x, y \in \mathcal{U}_y, \mathcal{U}_y^x \cap \mathcal{U}_y = \emptyset$$

Now, $\{\mathcal{U}_y \mid y \in K\}$ is open cover of K , therefore there is a finite subcover

$$\{\mathcal{U}_{y_i} \mid i=1, 2, \dots, r_x\} \text{ and open neighborhoods of } x \ \{\mathcal{U}_{y_i}^x \mid i=1, \dots, r_x\}.$$

Put $V_x = \bigcup_{i=1}^{r_x} \mathcal{U}_{y_i}^x$. Then, V_x is open neighborhood and $K \cap V_x = \emptyset$, thus $V_x \subseteq K^c$.

$$\text{(More specifically, } K \subseteq \bigcup_{i=1}^{r_x} \mathcal{U}_{y_i} \text{ and } \left(\bigcup_{i=1}^{r_x} \mathcal{U}_{y_i} \right) \cap V_x = \bigcup_{i=1}^{r_x} (\mathcal{U}_{y_i} \cap V_x) = \bigcup_{i=1}^{r_x} \emptyset = \emptyset.$$

And boundedness is clear.

Counterexample: Let $X = \{\frac{1}{n} \mid n \in \mathbb{N}\}$, discrete topology.

Then, it is closed itself, and bounded, but not Compact.

15. Let X be a Hausdorff, and $A, B \subseteq X$ be disjoint compact subsets.

Then, there exist disjoint open sets U and V containing A and B , respectively.

Proof. Let $x \in A$ be fixed. Then, for each $y \in B$, there exist opens $U_{(x,y)}$ and $V_{(x,y)}$ s.t.

$$x \in U_{(x,y)}, \quad y \in V_{(x,y)}, \quad U_{(x,y)} \cap V_{(x,y)} = \emptyset.$$

Since $\{V_{(x,y)} \mid y \in B\}$ is open cover of B , there exists a finite subcover

$$\{V_{(x,y_i)} \mid i=1, 2, \dots, r_x\}. \text{ Now, } U_x = \bigcap_{i=1}^{r_x} U_{(x,y_i)} \text{ is open neighborhood of } x.$$

$$V_x = \bigcup_{i=1}^{r_x} V_{(x,y_i)} \text{ is open set containing } B, \text{ and } U_x \cap V_x = \emptyset. \text{ because}$$

$$U_x \cap V_x = U_x \cap \left(\bigcup_{i=1}^{r_x} V_{(x,y_i)} \right) = \bigcup_{i=1}^{r_x} (U_x \cap V_{(x,y_i)}) = \bigcup_{i=1}^{r_x} \emptyset = \emptyset$$

In summary, for each $x \in A$, there are two disjoint opens U_x, V_x s.t.

$$x \in U_x, \quad B \subseteq V_x, \quad U_x \cap V_x = \emptyset.$$

And, $\{U_x \mid x \in A\}$ is an open subcover of A , thus it has a finite subcover

$$\{U_{x_i} \mid i=1, \dots, m\}. \text{ Finally,}$$

$$\bigcup_{i=1}^m U_{x_i} \text{ and } \bigcap_{i=1}^m V_{x_i} \text{ are disjoint open sets containing } A \text{ and } B, \text{ respectively.}$$

$$\text{Disjointness is given by } \left(\bigcup_{i=1}^m U_{x_i} \right) \cap \left(\bigcap_{i=1}^m V_{x_i} \right) = \bigcup_{i=1}^m \left(U_{x_i} \cap \left(\bigcap_{i=1}^m V_{x_i} \right) \right) = \bigcup \emptyset = \emptyset.$$

16. If $f: X \rightarrow Y$ is continuous map where X is compact and Y is Hausdorff, then f is closed map.

Proof. Let $C \subseteq X$ be a closed set. Since X is compact, C is compact.

And, the continuous map f preserves compactness, $f[C]$ is compact in T_2 , thus closed.

17. If Y is compact, then the projection $p_X: X \times Y \rightarrow X$ is a closed map.

Proof. Let $C \subseteq X \times Y$ be a closed set. Want to show that $X \setminus p_X[C]$ is open.

Let $x \in X \setminus p_X[C]$ be given. Then, $\{x\} \times Y \cap C = \emptyset$, therefore $\{x\} \times Y \subseteq (X \times Y) \setminus C$.

Now, by the Tube Lemma there is an open $U \subseteq X$ s.t.

$$\{x\} \times Y \subseteq U \times Y \subseteq (X \times Y) \setminus C.$$

Now, $x \in U$ and $U \subseteq X \setminus p_X[C]$, thus $X \setminus p_X[C]$ is open.

Note: the converse is true, but so hard to prove.

Recall: The Tube Lemma.

Let X be a topological space, and Y be a compact space.

Then, for the product space $X \times Y$, for fixed $x_0 \in X$,

For any open $N \subseteq X \times Y$ s.t. $\{x_0\} \times Y \subseteq N$, there is open $W \in \mathcal{C}_X$ s.t. $\{x_0\} \times Y \subseteq W \times Y \subseteq N$.

Proof. Since $\{x_0\} \times Y \rightarrow Y: (x_0, y) \mapsto y$ is homeo, $\{x_0\} \times Y$ is compact.

Since N is open, for any $y \in Y$, $\exists U_y \in \mathcal{C}_X, V_y \in \mathcal{C}_Y$ s.t. $\{x_0\} \times Y \subseteq U_y \times V_y \subseteq N$.

Now, open cover $\{U_y \times V_y \mid y \in Y\}$ of $\{x_0\} \times Y$ has a finite subcover

$$\{U_{y_i} \times V_{y_i} \mid i=1, \dots, N\}.$$

Set $W = \bigcap_{i=1}^N U_{y_i}$. Then, $\{x_0\} \times Y \subseteq W \times Y$ is clear, and

$$(x, y) \in W \times Y \Leftrightarrow \forall i=1, \dots, N, x \in U_{y_i} \text{ and } \exists y \in V_{y_i}$$

$$\Rightarrow (x, y) \in U_{y_i} \times V_{y_i} \subseteq N.$$

8. Let X be any Space, Y be a Compact Hausdorff. Then,

$f: X \rightarrow Y$ is continuous if and only if $G_f = \{(x, f(x)) \mid x \in X\} \subseteq X \times Y$ is closed.

Proof. Suppose that f is continuous. Let $(x, y) \in (X \times Y) \setminus G_f$ be given. Clearly $y \neq f(x)$. Since Y is T_2 , there exist disjoint opens $U, V \subseteq Y$ containing $y, f(x)$, respectively.

In other words, $(y, f(x)) \in U \times V \subseteq (Y \times Y) \setminus \Delta(Y)$

And $f(x) \in V$ implies $x \in f^{-1}[f[x]] \subseteq f^{-1}[V] \subseteq X$. Now enough to show that only (x, y)
 $(x, y) \in f^{-1}[V] \times U \subseteq (X \times Y) \setminus G_f$

because f is continuous, thus $f^{-1}[V]$ is open. The inclusion (*) equals to:

$$(f^{-1}[V] \times U) \cap G_f = \emptyset$$

and it is true, if it is not empty set, then for some $(x, f(x)) \in G_f$,
 $(x, f(x)) \in f^{-1}[V] \times U$. That is, $f(x) \in V$ and $f(x) \in U$, which is contradiction.

(Note: in this direction we used only T_2 Condition.)

Conversely, suppose that the graph G_f is closed set.

Let $U \subseteq Y$ be an open set. Want to show that: $f^{-1}[U]$ is an open set of X .

Let $x \in f^{-1}[U]$ be given. Then, $f(x) \in U$ by definition

Let $x_0 \in X$ be given. Then, for any open $V \subseteq Y$ containing $f(x_0)$,

$$G_f \cap (X \times (Y \setminus V))$$

is closed set. Now, since Y is Compact, the Projection $\pi_X: X \times Y \rightarrow X$ is closed map, thus

$$\pi_X[G_f \cap (X \times (Y \setminus V))] \subseteq X \text{ is closed in } X.$$

Now, $U = X \setminus \pi_X[G_f \cap (X \times (Y \setminus V))]$ is open, moreover, $x_0 \in U$ because

$$x_0 \in U \Leftrightarrow x_0 \notin \pi_X[G_f \cap (X \times (Y \setminus V))] \Leftrightarrow \nexists (x, y) \in G_f \cap (X \times (Y \setminus V)), \pi_X(x, y) = x \neq x_0$$

But, since $(x_0, f(x_0)) \notin (X \times (Y \setminus V))$, the last assertion holds. Now, only showing $f^{-1}[U] \subseteq V$ is remained.

Let $x \in U$. Then, $f(x) \notin Y \setminus V$, that is, $f(x) \in V$.

79. The Generalized Tube Lemma.

Let $A \subseteq X$, $B \subseteq Y$ be compact subspaces. Then,

For any open $N \subseteq X \times Y$ containing $A \times B$, there exists opens $U \subseteq X$ and $V \subseteq Y$ s.t. $A \times B \subseteq U \times V \subseteq N$

Proof.

72. Let $p: X \rightarrow Y$ be a closed continuous onto map s.t. for all $y \in Y$, $p^{-1}(\{y\})$ is compact.

If Y is compact, then X is compact. (Such p is called perfect map.)

Proof. Let $\mathcal{C} \subseteq \mathcal{C}_X$ be an open cover of X . That is, $\bigcup_{U \in \mathcal{C}} U = X$

Now, by the surjectivity, $p\left[\bigcup_{U \in \mathcal{C}} U\right] = \bigcup_{U \in \mathcal{C}} p[U] = Y$.